Homework: Server Data, Caching & Firebase Authentication

Extend your previous React Router project. Keep all existing functionality and add server-data caching, environment variables, and Firebase Authentication.

Tasks

Environment Variables & Firebase Setup

- ☐ Create a Firebase project and enable Email/Password and Google authentication.
- ☐ Store Firebase configuration in `.env.local` using `VITE_` variables.
- ☐ Initialize Firebase in a separate `firebase.js` file.
- ☐ Add `.env.local` to `.gitignore` and provide `.env.example`.

Authentication & Context

- ☐ Implement `AuthContext` with `AuthProvider` and `useAuth`.
- ☐ Track authentication using `onAuthStateChanged`.
- ☐ Support persistent sessions, loading states, and authentication errors.
- ☐ Implement signup, login, Google sign-in, and logout.

Authentication Pages & Routing

- ☐ `/login` — email/password login and Google sign-in.
- ☐ `/signup` — email, password, confirm password, and validation.
- ☐ `/profile` — protected page displaying the user's email and UID, with logout.
- ☐ Redirect unauthenticated users to `/login`.
- ☐ Update the NavBar to display Login/Signup or Profile/Logout depending on authentication state.

Server Data & Caching

- ☐ Implement caching for at least one API resource using a custom service or TanStack Query.
- ☐ Cache entries must use appropriate query keys and have an expiration or revalidation strategy.
- ☐ Invalidate the cache after relevant create/update/delete operations.
- ☐ Handle loading, error, and empty states.
- ☐ Prevent outdated requests from overwriting newer results.

Architecture

- ☐ Separate API, authentication, and caching logic into services or custom hooks.
- ☐ Do not access `localStorage` or make API requests directly inside UI components.
- ☐ Preserve all routes, filters, pagination, and functionality from the previous homework.

Acceptance Criteria

- ☐ Firebase authentication works with Email/Password and Google.
- ☐ Sessions persist after refresh.
- ☐ Protected routes and conditional navigation work correctly.
- ☐ Environment variables are configured.
- ☐ API caching and invalidation work. All previous project functionality remains operational.

Submission

1. GitHub repository link with updated README and `.env.example`.
2. Deployed application link
3. Short explanation in README describing the caching strategy and when cached data is invalidated.

References:

https://firebase.google.com/docs/auth/web/start

https://firebase.google.com/docs/auth/web/google-signin

https://firebase.google.com/docs/auth/web/auth-state-persistence

https://vite.dev/guide/env-and-mode

https://tanstack.com/query/latest/docs/framework/react/guides/caching